Lab Report: Text classification with neural networks

1 Introduction

In this report we compare several models on standard benchmarks. Each experiment was repeated 5 times with different random seeds. All experiments were run on a single GPU with 24 GB of memory.

2 Results

Figure 1 shows the validation accuracy of GRU by dataset.

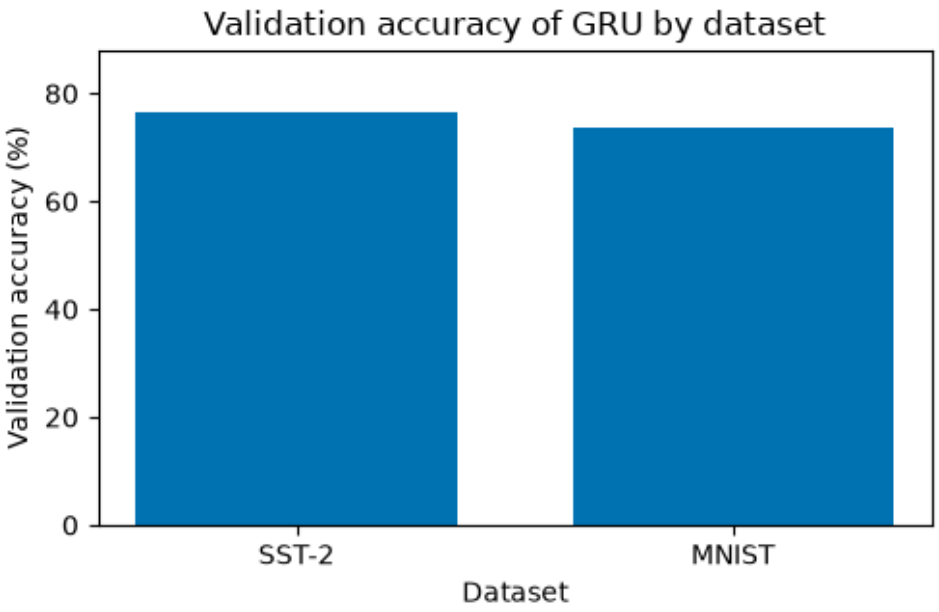


Figure 1: Validation accuracy of GRU by dataset for the models in this experiment.

As shown in Figure 1, on SST-2, GRU obtains a validation accuracy of 72.0%. GRU scores 20.5 points higher on SST-2 than on MNIST. On MNIST, GRU obtains a validation accuracy of 74%.

Figure 2 shows the validation accuracy of EfficientNet by dataset.

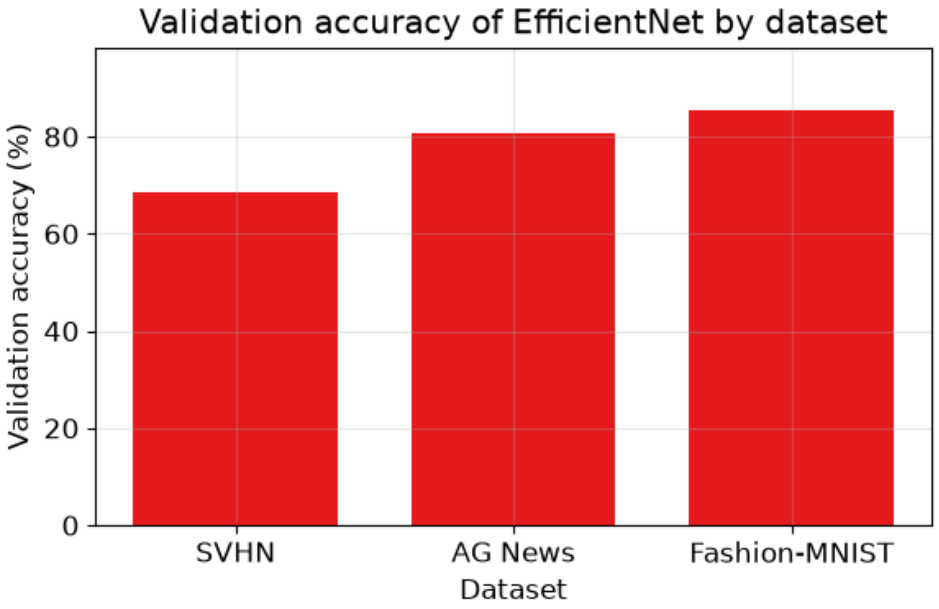


Figure 2: Validation accuracy of EfficientNet by dataset for the models in this experiment.

EfficientNet scores 17.2 points higher on Fashion-MNIST than on SVHN. On SVHN, EfficientNet obtains a validation accuracy of 88.1%.

Figure 3 shows the accuracy of MobileNet by dataset.

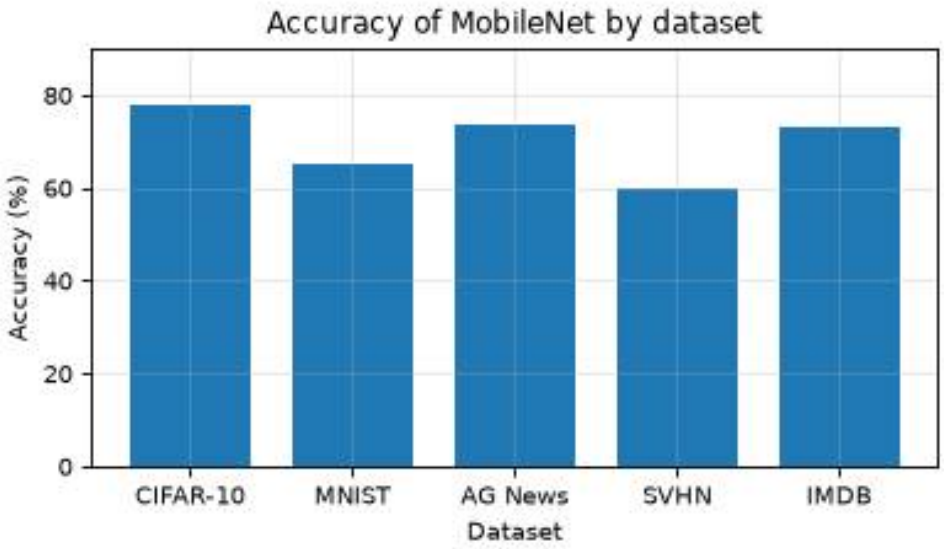


Figure 3: Accuracy of MobileNet by dataset for the models in this experiment.

MobileNet scores 53.2 points higher on AG News than on MNIST. MobileNet reaches 73% accuracy on IMDB.

3 Discussion

On IMDB, Logistic Regression obtains an accuracy of 80.3%. We note that the training was stable for all models.